

MATH 2130 - Week 1

Systems of Linear Equations, Vectors, and Matrices

1 Overview

The first half of this course is concerned with finding solutions to the **matrix equation**

$$Ax = b$$

Seems simple, doesn't it? And in a sense, it is. But as we'll see, it leads to a lot of very deep and beautiful mathematics.

We will start by considering systems of linear equations. Our primary tool in solving them will be **Gaussian Elimination**. From here, we will introduce **vectors** and **matrices**, at first as a means of simplifying the notation but then seeing how they simplify the computations as well. **Matrix multiplication** will be introduced, as will the multiplication of a vector and a matrix. Once we have these ideas under our belt, we will talk about **matrix factorizations**. This will lead naturally to **upper** and **lower triangular matrices** and **diagonal matrices**. Continuing with our exploration of matrices and their properties, we will look at **elementary matrices**, and also the important idea of the **inverse** of a matrix.

On a more computational level, we will discuss **row operations** leading to the **echelon** and row-reduced echelon form of a matrix and how it tells us very useful information about the solutions to a system of linear equations. We will see how these ideas connect to elementary matrices and to matrix factorizations, specifically into triangular forms.

The powerful open-source mathematical software program Sage will be introduced and the basics of using Sage will be covered. This will allow us to work with somewhat larger systems of equations and matrices without needing to spend too much time on very tedious computations.

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) **No Class!**
- (T) Systems of Linear Equations. Gaussian Elimination. Matrices and Vectors.
- (W) Matrix Multiplication. Triangular Matrices. Elementary Row Operations. Echelon Form.
- (R) Elementary Matrices. Matrix Factorization.
- (F) **No Class!**

Here is a checklist of things to do this week. This is meant to help you keep track of what you've completed and what you still need to work on.

- ☐ Go to the Google Drive folder for the class

<https://drive.google.com/drive/folders/1gcQUgDHnvywXIs6hv3JwfM-vsx0Us6hW>

and take a look around.

- ☐ Read these notes and ask any questions that you have about anything.
- ☐ Read the sections from Austin and Olver.
- ☐ Complete all of the homework exercises.

3 Key Concepts

- Systems of Linear Equations
- Gaussian Elimination (and Back Substitution)
- Vectors and Matrices
- Matrix Multiplication
- Triangular and Diagonal Matrices; Identity Matrices
- Symmetric Matrices; Transposes
- Elementary Row Operations and Elementary Matrices
- Augmented Matrices
- Echelon and Row-Reduced Echelon Form
- Nonsingular Matrices and Inverses
- Matrix Factorizations

4 Important Definitions, Formulas, and Theorems

In all the matrices that follow in this section, all the empty entries are 0's.

- An $m \times n$ **rectangular matrix** A has m rows and n columns of **entries**, which for us will almost always be real numbers, though occasionally they will be complex numbers (in general, they can be any element from a field). The entries of A are denoted by a_{ij} where the first index tells you the row that the entry is in, while the second index tells you the column that the entry is in. We will denote matrices by capital letters, like $A, B, D, L, U, \dots$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \cdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

- If the number of rows equals the number of columns, then the matrix is said to be a **square matrix**.
- Let A be an $m \times n$ matrix with entries a_{ij} . The **transpose** of A , denoted A^T , is the matrix that has entries a_{ji} . It is an $n \times m$ matrix. To get A^T , interchange all rows and columns of A :

$$A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ a_{13} & a_{23} & \cdots & a_{n3} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nm} \end{bmatrix}$$

- The entries a_{ii} in a square matrix are called the **main diagonal**.
- An $n \times n$ diagonal matrix has all non-diagonal entries equal to 0: $a_{ij} = 0$ if $i \neq j$, $a_{ii} = d_i$. The d_i can equal anything (including 0).

$$D = \begin{bmatrix} d_1 & & & & \\ & d_2 & & & \\ & & d_3 & & \\ & & & \ddots & \\ & & & & d_{n-1} & \\ & & & & & d_n \end{bmatrix}$$

The product of any number of diagonal matrices is diagonal: $D_1 D_2 \cdots D_k = D$.

The product DA of a diagonal matrix and any $n \times n$ matrix A multiplies row i of A by d_i , for each $1 \leq i \leq n$.

The product AD of a diagonal matrix and any $n \times n$ matrix A multiplies column j of A by d_j , for each $1 \leq j \leq n$.

- The $n \times n$ identity matrix, I_n , is an $n \times n$ diagonal matrix with all $d_i = 1$.

$$I_n = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix}$$

The identity matrix times any $n \times n$ matrix A always gives A : $I_n A = A I_n = A$.

- An upper triangular matrix, U , has all entries below the main diagonal equal to 0: $a_{ij} = 0$ if $i > j$.

$$U = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ & a_{22} & a_{23} & \cdots & a_{2n} \\ & & a_{33} & \cdots & a_{3n} \\ & & & \ddots & \vdots \\ & & & & \ddots & \vdots \\ & & & & & a_{mn} \end{bmatrix}$$

The product of any number of upper triangular matrices is upper triangular: $U_1 U_2 \cdots U_k = U$.

- An lower triangular matrix, L , has all entries above the main diagonal equal to 0: $a_{ij} = 0$ if $i < j$.

$$L = \begin{bmatrix} a_{11} & & & & \\ a_{21} & a_{22} & & & \\ a_{31} & a_{32} & a_{33} & & \\ \vdots & \vdots & \vdots & \ddots & \\ \vdots & \vdots & \vdots & & \ddots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & \cdots & a_{mn} \end{bmatrix}$$

The product of any number of lower triangular matrices is lower triangular: $L_1 L_2 \cdots L_k = L$.

- A square matrix A is symmetric is $a_{ij} = a_{ji}$ for all $1 \leq i, j \leq n$.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{12} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{13} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{nn} \end{bmatrix}$$

Any symmetric matrix is equal to its own transpose: $A^T = A$.

- Let A and B be two $m \times n$ matrices and let c be any scalar. Then

$$A+B = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \cdots & a_{1n}+b_{1n} \\ a_{21}+b_{21} & a_{22}+b_{22} & \cdots & a_{2n}+b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}+b_{m1} & a_{m2}+b_{m2} & \cdots & a_{mn}+b_{mn} \end{bmatrix}, \quad \text{and} \quad cA = \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1n} \\ ca_{21} & ca_{22} & \cdots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \cdots & ca_{mn} \end{bmatrix}$$

- An $m \times 1$ matrix is called a **column vector**. The entries of v will be written as v_i with $1 \leq i \leq m$:

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{bmatrix}$$

The transpose of a column vector is called a **row vector**. We will denote vectors by lower-case letters, like $b, c, v, w, x, \dots$

- Unless otherwise specified, all vectors will be column vectors.
- The **inner product** of the vectors v and w (both with m entries) is defined to be $v^T w$, with

$$v^T w = \begin{bmatrix} v_1 & v_2 & \cdots & v_m \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix} = v_1 w_1 + v_2 w_2 + \cdots + v_m w_m$$

This will be written as either $v \cdot w$ or $\langle v, w \rangle$. It is a **scalar** (which for us will be either a real number or a complex number).

In words, to form the inner product of two vectors, we write the first as a row vector and the second as a column vector, and we multiply the entries across the row vector and down the column vector, and add the resulting products.

- If A is an $m \times n$ matrix with rows given by $r_1, r_2, \dots, r_m$ and B is an $n \times p$ matrix with columns $c_1, c_2, \dots, c_p$, the **matrix product** AB is the $m \times p$ matrix with ij -entry equal to the inner product of row i of A with column j of B :

$$AB = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} \begin{bmatrix} c_1 & c_2 & \cdots & c_p \end{bmatrix} = \begin{bmatrix} r_1 \cdot c_1 & r_1 \cdot c_2 & \cdots & r_1 \cdot c_p \\ r_2 \cdot c_1 & r_2 \cdot c_2 & \cdots & r_2 \cdot c_p \\ \vdots & \vdots & \ddots & \vdots \\ r_m \cdot c_1 & r_m \cdot c_2 & \cdots & r_m \cdot c_p \end{bmatrix}$$

In words, the product of an $m \times n$ matrix A and an $n \times p$ matrix B is found by “multiplying across the rows of A and down the columns of B .” Please note that the number of columns of the first matrix must equal the number of rows of the second matrix. In this case, we say that the size of the matrices are **compatible**.

- In general, even if the size of the matrices A and B are compatible for both AB and BA to be defined, $AB \neq BA$. This means that matrix multiplication is **NOT** commutative. The order matters!
- Matrix multiplication is associative: if three matrices A, B, C are compatible, then $(AB)C = A(BC)$.
- Matrix multiplication is distributive: if the matrices A, B, C, D are compatible, then $A(B + C) = AB + AC$ and $(B + C)D = BD + CD$. Please note the order of the products and remember that matrix multiplication is not commutative, so you can’t change AB to BA , for example.
- Let A be a $n \times n$ square matrix. If there exists another $n \times n$ matrix B such that $AB = BA = I_n$, then B is said to be the **inverse** of A and is written A^{-1} . So we have $AA^{-1} = A^{-1}A = I_n$. If the inverse exists, it is unique.
- If A and B are invertible $n \times n$ matrices, then $(AB)^{-1} = B^{-1}A^{-1}$.
- There are three **elementary row operations**:
 1. Multiplying any row by a non-zero scalar.
 2. Exchanging any two rows.
 3. Adding a scalar multiple of any row to another row.

Elementary row operations correspond to multiplying by **elementary matrices**.

$$\text{multiplying row } i \text{ by } c = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & c & \\ & & & & 1 \end{bmatrix} \quad (\text{the } c \neq 0 \text{ is in row } i)$$

$$\text{exchanging row } i \text{ and row } j = \begin{bmatrix} 1 & & & \\ & & & 1 \\ & & \ddots & \\ & 1 & & \\ & & & & 1 \end{bmatrix} \quad (\text{the 1's in row } i \text{ and } j \text{ switched})$$

$$\text{adding } c \text{ times row } i \text{ to row } j = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & c & \\ & & & & 1 \end{bmatrix} \quad (\text{the } c \text{ is in row } j, \text{ column } i)$$

We will focus primarily on row operations of the third type: adding a multiple of row i to row j . We will denote these types of elementary matrices by E_{ji} . Please note the order of the indices. The first index is the row that is to be changed, while the second indicates the row that is multiplied by a constant. Thus, E_{21} means that the second row of the matrix is to be changed by adding a multiple of the first row of the matrix. We do not explicitly mention the value of the scalar in the indices.

- All elementary matrices are invertible.
- The importance of elementary matrices is that they allow us to use matrix multiplication to perform Gaussian elimination. This will be easiest to see in an example. So we will look at an example here, even though this is not the Examples section.

Let the system of equation $Ax = b$ be written as the augmented matrix $[A|b]$ below:

$$[A|b] = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 4 & 1 & 0 & -2 \\ -2 & 2 & 1 & 7 \end{array} \right]$$

be the augmented matrix that corresponds to the system of equations listed at the beginning of the Examples section. We want to use the (1,1)-entry (called a **pivot**) to eliminate the values that lie below it in the first column. We will do this in two steps: first, by adding -2 times the first row to the second row; second, by adding the first row to the third row. The elementary matrices that correspond to this are given by (remember our indexing notation):

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Then we have

$$E_{21}A = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 0 & -1 & -2 & -4 \\ -2 & 2 & 1 & 7 \end{array} \right]$$

and

$$E_{31}(E_{21}A) = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 0 & -1 & -2 & -4 \\ 0 & 3 & 2 & 8 \end{array} \right]$$

We have “cleared out” all the non-zero values in the first column below the pivot in the (11)-entry.

The next pivot is the -1 in the (22)-entry. Now we will use

$$E_{32} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{array} \right]$$

to remove the 3 in the second column below the pivot:

$$E_{32}(E_{31}E_{21}A) = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 0 & -1 & -2 & -4 \\ 0 & 0 & -4 & -4 \end{array} \right] = [U|c]$$

The thing to note here is that the transformed matrix on the left of the vertical line is upper triangular. Call it U . So these three elementary row operations have transformed the system $Ax = b$ into an equivalent system $Ux = c$, where U and c are as above.

We see that the relationship between A and U and b and c are completely determined by the three elementary matrices E_{21}, E_{31}, E_{32} :

$$E_{32}E_{31}E_{21}A = U, \quad \text{and} \quad E_{32}E_{31}E_{21}b = c$$

We want to then look at the matrix $E_{32}E_{31}E_{21}$:

$$E_{32}E_{31}E_{21} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & 3 & 1 \end{array} \right]$$

Notice that this matrix is lower triangular and it is invertible, since each elementary matrix is invertible. Inverting these matrices and forming the product gives

$$L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{array} \right]$$

We can now get to the main point of all of this. Since $(E_{32}E_{31}E_{21})A = U$ and we have that $(E_{32}E_{31}E_{21})^{-1} = L$, we can multiply both sides of $(E_{32}E_{31}E_{21})A = U$ by L to get that $A = LU$:

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 1 & 0 \\ -2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & -4 \end{bmatrix} \quad (\text{check this!})$$

So we have factored the matrix A into the product LU of a lower triangular and upper triangular matrices! This will always be possible, as long as none of the pivots that arise in the process of Gaussian elimination are zero.

- As long as no pivots are zero, the matrix A can be written as the product LU of a lower triangular matrix L and an upper triangular matrix U .

The entries of L on the main diagonal are 1's; below the diagonal, they are the multipliers l_{ij} of row j which are subtracted from row i during elimination. (Watch the subscripts!)

U is the coefficient matrix which appears after elimination and before back-substitution; its diagonal entries are the pivots.

- Returning to the previous example, since all the pivots of A are assumed to be non-zero, all the entries on the main diagonal of U are non-zero (since they are the pivots). This means that we “divide them out” in the following way:

$$U = \begin{bmatrix} 2 & & \\ & -1 & \\ & & -4 \end{bmatrix} \begin{bmatrix} 1 & 1/2 & 1/2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

This gives us a further factorization, this time of U , into the product of a diagonal matrix and another triangular matrix: $U = DU_1$.

- As long as no pivots are zero, the matrix A can be written as the product LDU where L is the same as in the previous factorization, D is the diagonal matrix with pivots d_i along the diagonal, and U is upper triangular with 1's on the main diagonal and

$$U = \begin{bmatrix} 1 & u_{12}/d_1 & u_{13}/d_1 & \cdots \\ & 1 & u_{23}/d_2 & \cdots \\ & & \ddots & \vdots \\ & & & 1 \end{bmatrix}$$

- If A is symmetric, and if it can be factored into $A = LDU$ without using any row exchanges, then the upper triangular U is the transpose of the lower triangular L : $A = LDL^T$.

5 Examples

- We will spend a lot of time this week working with the following system of linear equations:

$$\begin{array}{rrcrcl} 2u & + & v & + & w & = & 1 \\ 4u & + & v & & & = & -2 \\ -2u & + & 2v & + & w & = & 7 \end{array}$$

and the associated augmented matrix

$$A = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 4 & 1 & 0 & -2 \\ -2 & 2 & 1 & 7 \end{array} \right]$$

- We will also work with the following **tridiagonal** matrix (where all the empty entries are 0's):

$$F = \begin{bmatrix} 2 & -1 & & & & & \\ -1 & 2 & -1 & & & & \\ & -1 & 2 & -1 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & -1 & 2 & -1 & \\ & & & & -1 & 2 & -1 \\ & & & & & -1 & 2 \end{bmatrix}$$

- The symmetric matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 8 \end{bmatrix}$$

factors into LDL^T in the following way:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad (\text{check this!})$$

The pivots of the original system are 1, 4 and the first and last matrices on the right-hand side are transposes (the first is lower triangular and the third is upper triangular).

6 Sage Code

6.1 Commands

These are the most common Sage commands that we will use this week. Think of this more as a reference. It is usually easier to understand what the commands do by seeing them used in example code, which is given below.

- `%display latex`

This is a formatting command that tells Sage to display all output in the `displaymath` style of latex. It just makes things nicer to look at and easier to read.

- `matrix([[a11, a12, ..., a1n], [a21, a22, ..., a2n], ..., [am1, am2, ..., amn]])`

or

`matrix(m, n, [a11, a12, ..., amn])`

These commands will generate an $m \times n$ matrix with the mn entries $a_{11} \dots, a_{mn}$. If $n = 1$, this will create a **column vector** with m entries. If $m = 1$, it will create a **row vector** with n entries.

- `.rref()`

If this command is appended to a matrix A , it will return the row-reduced echelon form of A . We will use this a lot.

- `.echelon_form()`

If this command is appended to a matrix A , it will return the echelon form of A . This will give an upper-triangular matrix, whereas `.rref()` will try to reduce things to a diagonal matrix. Sometimes the echelon form will contain more information than the rref form.

- `.echelon_form(transformation = True)`

This will give the echelon form and also return the product of elementary matrices (call this matrix E) that transform the original matrix into the echelon form. If the original matrix is A , then `A.echelon_form(transformation = True)` will return two matrices, R and E such that R is the echelon form of A and we have $EA = R$.

- `import sympy as sy`

Sometimes Sage tries to do too much. It will usually try to reduce everything as far as possible, which isn't always what we want. There is a Python package, `sympy` (which stands for SYMBolic PYthon) which allows us more control over what Sage does. To load the package, include the line above at the beginning of the cell. Then you can call commands from the package by prepending `sy.` before the command name. See the examples below.

- `.solve_right()`

If you have a matrix A and a vector b , the command `A.solve_right(b)` will find the solution to $Ax = b$ (which is the vector x) if a solution exists, and generate an error message if a solution doesn't exist.

- `.LU()`

Will calculate the LU factorization of the matrix it is appended to. Please note: this command will produce a triple $[P, L, U]$, such that $PA = LU$. Here P is a permutation matrix (which is rearranging the rows of A). If no row exchanges are necessary, then $P = I$, the identity matrix.

6.2 Example Code

- We'll start with something simple:

```
% display latex
A = Matrix(4,5,[1, 3, 2, -1, 0, 2, 6, 1, 4, 3, -1, -3, -3, 3, 1, 3, 9, 8, -7, 2])
A.echelon_form(), A.rref()
```

The first line is telling Sage to display things as if they were typeset in latex, which makes the resulting output easier to read. The second line is defining the 4×5 matrix A . The third line is showing both the echelon form and the row-reduced echelon form of the matrix A . These are very similar but they are different in the fifth column.

- Sometimes we want to work with undetermined constants within our matrices. For example, we might want to know what values of the constants a, b, c will ensure that the following matrix is nonsingular:

$$A = \begin{bmatrix} 1, 2, a \\ 0, 1, b \\ 3, 9, c \end{bmatrix}$$

What we want to do is use elementary row operations to reduce the matrix A to echelon form and have the third column have entries that are in terms of the constants a, b, c . There's no easy way to do this directly in Sage, but we can use the following code to do it using `sympy`

```

import sympy as sy
sy.init_printing()
a, b, c = sy.symbols('a b c ', real = True, constant = True)
A = sy.Matrix(3,3,[1,0,a,0,1,b,3,9,c])
A.echelon_form()

```

The first line is loading the sympy package and abbreviating is as `sy` (this means we only have to type `sy.` instead of `sympy.` when we want to call commands from the sympy package). The second line prints the results in an easier to read format. It is not necessary, but it makes the output **MUCH** easier to read. You should use it. The third line is defining a, b, c as symbols that are real-valued constants. The fourth line is creating the matrix A . Please note that you must capitalize the M in `Matrix`. The fifth line is returning the echelon form of A . The third column will be in terms of the variables a, b, c , which is what we want. Don't use the `.rref()` command here. This will reduce things down to the identity matrix, which isn't what we want.

- This code defines the matrix B and computes the echelon form. The option `transformation = True` returns the matrix E such that $EB = R$, where R is the echelon form of B . This can be a very useful matrix to have.

```

% display latex
B = Matrix([[1,2,1],[0,1,2],[3,9,0]])
R = B.echelon_form(transformation = True)[0]
E = B.echelon_form(transformation = True)[1]
R, E, E*B

```

The first line is telling Sage to output the display as if it were typeset in latex. This makes it much easier to read. The second line is defining the matrix. The `.echelon_form(transformation = True)` command returns a list of two matrices, $[R, E]$, where R is the echelon form and E is the transformation matrix (the product of all the elementary matrices transforming B into R). The third line is defining R to be the first item of this list (please **NOTE**: lists are indexed starting from 0 in Sage). The fourth line is defining E to be the second item of the list. The fifth line is showing R , E , and computing EB (which is the same as R).

- This code finds the echelon form of the augmented matrix $[A|s]$ which can be used to find the solution space to the system $Ax = s$. Note that this is a system of 4 equations in 5 unknowns and it would be very tedious to solve this by hand.

```

import sympy as sy
sy.init_printing()
a,b,c,d = sy.symbols('a b c d', real = True, constant = True)
A = sy.Matrix(4,5,[1, 3, 2, -1, 0, 2, 6, 1, 4, 3, -1, -3, -3, 3, 1, 3, 9, 8, -7, 2])
s = sy.Matrix(4,1,[a,b,c,d])
B = A.row_join(s)
B.echelon_form()

```

The two things to note here are: line four is creating the column vector of constants a, b, c, d ; line five is creating the augmented matrix by using the `.row_join()` command.

7 Additional Comments

- It is hard to over-emphasize just how important solving linear systems is. It arises in all sorts of problems, particularly in applied mathematics and in any field that uses applied mathematics in any

meaningful way. Most of the systems of equations that we will look at this semester will be very small. This is not the case in the “real world.” It is not uncommon for the systems in real-world applications to have thousands of equations in thousands of unknowns. In these situations, computers must be used and used efficiently. We will not go into any of the practical matters involved in solving such huge systems, but they involve cutting-edge mathematics, sophisticated algorithms, and a deep understanding of both the hardware and software abilities of the technology being used.

- It is also hard to over-emphasize just how important Gaussian Elimination is, both as a conceptual and computational tool. It is the foundation of a huge part of numerical linear algebra and so underlies much of applied mathematics and its use in related fields.
- Matrices are incredible things. They live in-between numbers and functions, and can be thought of as higher-dimensional numbers and discrete (and so finite) approximations of continuous functions. They are extremely flexible objects that have come to dominate much of modern mathematics. They are ubiquitous and new uses for them are being found all of the time. On the other hand, it is worth keeping in mind that they are still relatively “young” mathematical objects, only about 170 years old. The rise of importance of matrices is intimately connected with the rise of computational technology. It is safe to say that linear algebra as we know it wouldn’t exist without the simultaneous existence of powerful computers.
- Connected to the previous comment, you will **NEVER** do any real matrix computation by hand. It just won’t happen. So the sooner you can get comfortable using technology, the better. Please take Sage seriously. You can thank me later.

8 Reading Notes

The best resource for this material is the pdf of the material from Peter Olver’s book *Applied Linear Algebra*. I have included the entire pdf of this book in the Google Drive folder. I have also broken it into separate pdfs for each week of the course. You should start with the weekly pdf, but if you can’t find what you’re looking for, then look in the whole-book pdf. I wouldn’t recommend doing a general Google search since you will get way too many hits and things will be very difficult to sort through.

For each section of Olver contained in the Week 1 pdf, I’ve indicated the main topics covered in that section. Use this to help guide your outside reading (and you should be doing outside reading).

- Olver 1.1 - Systems of Linear Equations
- Olver 1.2 - Vectors and Matrices, Matrix Multiplication
- Olver 1.3 - Gaussian Elimination, Elementary Matrices
- Olver 1.4 - Permutations and Permutation Matrices
- Olver 1.5 - Inverses
- Olver 1.6 - Transposes and Symmetric Matrices
- Olver 1.7 - Tridiagonal Matrices
- Olver 1.8 - Solving General Systems

9 Homework Exercises

Please do **ALL** exercises. Not all of them will be graded for correctness, but they will all be checked for completion. Failure to do all of them will lower the completion portion of the Homework score.

Please write your solutions up clearly and cleanly, using full English sentences. Try not to have a bunch of arrows pointing from one thing to another or a bunch of things crossed-out and a bunch of other things circled, etc. Presentation matters.

1. Find examples of 2×2 matrices such that

- (a) $A^2 = -I$, all entries of A are real numbers.
- (b) $B^2 = 0$, with $B \neq 0$.
- (c) $CD = -DC$, with $CD \neq 0$.
- (d) $EF = 0$, with no entries of E or F equal to 0.

2. Let

$$A = \begin{bmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 6 & 9 & 8 \end{bmatrix}$$

Factor A into LU .

3. In this question you will solve $Ax = b$ by performing two steps. The factors $A = LU$ and b are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$$

- (a) Do **forward elimination** by solving $Lc = b$ for the unknown vector c .
- (b) Do **back substitution** by solving $Ux = c$ for the unknown vector x .

4. Let

$$M = \begin{bmatrix} 0 & 0 & 0 & 1 & 2 \\ 1 & -5 & -1 & 0 & -3 \\ -2 & 10 & 3 & 5 & 17 \\ -2 & 10 & 3 & 2 & 11 \end{bmatrix}$$

be the augmented matrix of the linear system given by the matrix equation $Ax = b$, and let

$$R = \begin{bmatrix} 1 & -5 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

be the rref of M .

- (a) How many equations and how many variables does the linear system have?
- (b) Is the linear system consistent?
- (c) What are the pivot positions?
- (d) What are the basic and free variables?
- (e) Describe the solutions.
- (f) Describe the solutions to the homogeneous equation $Ax = \vec{0}$.

5. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 6 \\ 5 & 10 & 15 \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Is the system $Ax = b$ solvable? Please justify your answer.

6. Let

$$A = \begin{bmatrix} 1 & 4 & 3 & -5 & -11 & 10 \\ 0 & 0 & 1 & -1 & -3 & 3 \\ -1 & -4 & -5 & 7 & 17 & -16 \\ 1 & 4 & 2 & -4 & -8 & 7 \end{bmatrix}$$

and let

$$R = \begin{bmatrix} 1 & 4 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & -1 & -3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

be the rref of A .

- (a) What are the pivot columns of A ?
 - (b) What two vectors are guaranteed to span the column space of A ?
 - (c) All of the other column vectors in A can be written in terms of the vectors from the previous part. Write these columns (the four ones different from the answer you gave to the previous part) of A as linear combinations of the vectors from part (b).
7. Find the inverse of the matrix A by putting the augmented matrix $[A|I_3]$ into rref form: $[I_3|A^{-1}]$.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

8. Compute the LDL^T factorization of

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 12 & 18 \\ 5 & 18 & 30 \end{bmatrix}$$